Decisions Not Made: Owner Attention, the Intrinsic Value of Authority, and the Limits to Organizational Scale

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In the name of God, the Most Gracious, the Most Merciful.

Abstract

Owner-managed firms frequently stop growing well before technology, demand, or finance would require them to. This paper develops a span-of-control model in which the binding constraint is the owner's attention and the central choice is which decisions to stop making. Delegating a decision frees attention at the cost of an oversight requirement and a decision-quality loss. The optimal policy is a threshold rule: the owner delegates every decision whose delegation loss falls below the attention it frees, valued at the average return to the owner's time. Four results follow. First, behavioral sources of under-delegation (an intrinsic value of authority and a salience-driven overweighting of delegation errors) are equivalent to a reduction in the perceived value of the business, and the authority rent distorts behavior only to the extent that oversight is costly. Second, under-delegation is first order for scale but second order for profit, so the owner's own profit signal provides weak feedback against it. Third, when delegation capability accumulates with use and depreciates with disuse, the model admits multiple steady states; owners whose formative years were spent at a scale where delegation was unnecessary converge to a low-delegation trap. Fourth, in a recursive hierarchy the returns to delegation compound across tiers, so a uniform under-delegation culture imposes losses that grow geometrically with the depth of a business group. An accounting identity yields an observable diagnostic: sustained growth without neglected decisions requires the number of decisions the owner no longer makes to grow faster than the portfolio itself.

JEL codes: D23, D91, L22, L25, M12, G34

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1. Introduction

A recurring observation about owner-managed firms is that their growth is limited less by markets than by their owners. Firms in many economies remain small for long periods even when their owners are capable, their products are competitive, and external finance is available. Detailed field evidence attributes part of this pattern to the owner's reluctance or inability to pass decisions to others. In a well-known management experiment among Indian textile firms, owners cited their own limited time and a lack of trust in non-family managers as the reasons their firms did not expand (Bloom et al., 2013). Cross-country evidence relates decentralization to trust and links it to firm size (Bloom, Sadun, and Van Reenen, 2012), and recent quantitative work attributes a meaningful share of the gap in firm dynamics between developing and developed economies to limits on delegation (Akcigit, Alp, and Peters, 2021).

The canonical explanations for limited delegation are informational and contractual: agents are imperfectly aligned, contracts are incomplete, and institutions that would make outside managers reliable are weak. These explanations are important, but they leave a residual that is difficult to ignore. Experimental evidence shows that individuals attach intrinsic value to holding decision rights, over and above the instrumental value of the decisions themselves (Bartling, Fehr, and Herz, 2014; Owens, Grossman, and Fackler, 2014), and that authority affects behavior in ways that are not reducible to incentives (Fehr, Herz, and Wilkening, 2013). Psychological evidence documents a persistent tendency to overweight the value of personal control (Langer, 1975). And founders typically acquire their managerial habits at a scale at which delegation is unnecessary, so the practices that produced early success are formed under conditions that no longer hold once the firm grows.

This paper offers a tractable framework in which these forces can be analyzed jointly and compared with the standard instrumental motive. The framework builds on the span-of-control tradition (Lucas, 1978; Calvo and Wellisz, 1978; Rosen, 1982; Garicano, 2000) but shifts attention from the allocation of *talent* to the allocation of *decisions*. An owner endowed with a fixed amount of attention operates a portfolio of units, each of which generates a flow of decisions. Each decision can be made personally, which consumes a unit of attention, or delegated, which consumes a smaller amount of oversight attention and entails a decision-specific loss. Scale is then determined jointly by the owner's attention endowment and by the fraction of decisions the owner is willing to stop making.

The analysis yields five main results.

The first characterizes optimal delegation. The owner delegates according to a threshold rule, and the threshold has a simple interpretation: a decision is delegated whenever its delegation loss is smaller than the attention it frees, valued at the *average* product of the owner's attention. At small scale, when the owner's attention is not binding, the optimal policy is full control; delegation becomes valuable only once the attention constraint binds. The model thus formalizes a regime change: the managerial practice that is optimal for a single establishment is suboptimal for many.

The second result concerns behavioral sources of under-delegation. I introduce two departures from profit maximization: an intrinsic utility from personally exercised authority, and a salience distortion that overweights the visible losses from delegated decisions relative to the invisible opportunity cost of the owner's time. Both distortions turn out to be equiva-

lent to a reduction in the perceived value of the business, which yields sharp comparative statics. A less obvious implication is that the authority rent distorts delegation only to the extent that oversight of delegated decisions is costly. When oversight is free, the attention constraint already fixes the total number of decisions the owner can make personally, so the owner cannot consume more authority by delegating less. Improvements in monitoring technology therefore attenuate the behavioral distortion as well as the instrumental cost of delegation.

The third result is an asymmetry. Because the profit function is flat at the optimum while scale is strictly increasing in delegation, under-delegation produces first-order losses in scale but only second-order losses in profit. An owner who under-delegates runs a smaller organization that is nearly as profitable per unit of attention. The owner's own profit signal therefore supplies weak feedback against the habit, which helps explain why under-delegation can persist among owners who are otherwise disciplined by market outcomes.

The fourth result treats delegation as a capability. If the oversight cost of delegation declines with accumulated delegation experience and rises with disuse, the static model becomes a dynamic system with potentially multiple steady states. Owners whose early careers were spent in the small-scale regime, where full control is optimal, enter the growth phase with little delegation capital and may converge to a low-delegation steady state from which a myopic owner never escapes. Temporary shocks that force delegation, such as an owner's absence, can have permanent effects.

The fifth result extends the model to a recursive hierarchy, which is the natural representation of a business group or holding company. When each tier faces a delegation problem of the same form as the owner's, the gains from delegation multiply across tiers. A uniform behavioral distortion, interpreted as a managerial culture of under-delegation, therefore imposes losses that grow geometrically with organizational depth. The work of the owner at the apex consists almost entirely of decisions about the people who make decisions.

Finally, an accounting identity converts the model into a diagnostic. Holding the owner's attention fixed, the number of decisions the owner no longer makes must grow more than one-for-one with the number of units if the portfolio is to expand without decisions being neglected. Growth in the number of units that is not matched by growth in delegated decisions necessarily appears as a rising share of decisions that are neither made nor delegated. The number of firms an owner controls is thus a poor measure of organizational capacity; the number of decisions the owner has stopped making is a better one.

The remainder of the paper is organized as follows. Section 2 relates the paper to the literature. Section 3 presents the model and characterizes optimal delegation. Section 4 introduces behavioral distortions and derives the scale-profit asymmetry. Section 5 distinguishes delegation of objectives from prescription of solutions. Section 6 develops the dynamic model of delegation capability. Section 7 analyzes recursive hierarchies. Section 8 derives the diagnostic. Section 9 discusses empirical implications and measurement, and Section 10 concludes. Proofs are collected in Appendix A and details of the numerical illustrations in Appendix B.

2. Related Literature

The paper contributes to four strands of work.

Span of control and hierarchies. The idea that managerial capacity limits firm size goes back at least to Penrose (1959), and was formalized in models of hierarchical control and loss of control across layers (Williamson, 1967; Calvo and Wellisz, 1978), in the assignment of managerial talent to firm size (Lucas, 1978; Rosen, 1982), and in knowledge-based hierarchies in which problems are routed upward to those able to solve them (Garicano, 2000; Garicano and Rossi-Hansberg, 2006; see Garicano and Rossi-Hansberg, 2015, for a review). Related work studies hierarchies as structures for processing information under limited attention (Geanakoplos and Milgrom, 1991; Radner, 1993) and the allocation of organizational attention more generally (Dessein, Galeotti, and Santos, 2016). Relative to this literature, the present model takes the owner's attention as the scarce input and makes the delegation margin, rather than the assignment of talent, the object of choice. Its tractability allows behavioral distortions, capability dynamics, and recursive structures to be analyzed within a single framework.

The allocation of decision rights. A large literature studies when a principal should delegate authority to a better-informed but imperfectly aligned agent (Holmström, 1984; Aghion and Tirole, 1997; Dessein, 2002; Alonso, Dessein, and Matouschek, 2008). These models typically consider a single decision and ask how much discretion to grant. The present model abstracts from the micro-foundations of the delegation loss and instead considers a continuum of heterogeneous decisions competing for a fixed stock of attention. The loss from delegating a given decision can be interpreted as the reduced form of any of these models. Section 5 draws on the distinction between formal and real authority (Aghion and Tirole, 1997) and between general and specific knowledge (Jensen and Meckling, 1992) to analyze the difference between delegating an objective and prescribing a solution.

Behavioral organizational economics. Experimental work documents that individuals value decision rights intrinsically (Bartling, Fehr, and Herz, 2014), are willing to pay a premium to retain control (Owens, Grossman, and Fackler, 2014), and that the delegation of authority affects effort on both sides of the relationship (Fehr, Herz, and Wilkening, 2013). Evidence on executive overconfidence (Malmendier and Tate, 2005) and on the persistent influence of individual managers on corporate policy (Bertrand and Schoar, 2003) points in the same direction. The present paper embeds two such preferences in an equilibrium model of scale and shows that they have distinct comparative statics, in particular that the authority rent interacts with oversight costs in a specific way.

Firm growth, family firms, and business groups. The paper relates to empirical work on management practices (Bloom and Van Reenen, 2007), on the decentralization of decision making across countries (Bloom, Sadun, and Van Reenen, 2012), on managerial constraints in developing economies (Bloom et al., 2013; Akcigit, Alp, and Peters, 2021), on how CEOs allocate their time (Bandiera et al., 2018; Bandiera et al., 2020), and on family control (Bennedsen et al., 2007; Bertrand et al., 2008). It also relates to work on business groups and pyramidal ownership (Khanna and Yafeh, 2007; Almeida and Wolfenzon, 2006). Akcigit,

Alp, and Peters (2021) is the closest contribution: they embed an owner's time constraint and costly outside managers in a quantitative model of firm dynamics. The present paper is complementary. It is qualitative and theoretical, it focuses on the owner's preferences over authority and on the dynamics of delegation capability, and it derives implications for the internal structure of multi-tier groups.

3. A Model of Delegation under an Attention Constraint

3.1. Environment

An owner is endowed with $T > 0$ units of attention per period. The owner operates $N \geq 0$ identical units (establishments, subsidiaries, or companies), and the number of profitable units available is bounded by $M > 0$, which summarizes the size of the market or the stock of investment opportunities. Each unit generates a gross surplus $V > 0$ per period, net of all costs other than those associated with decision making.

Each unit generates a unit mass of decisions per period. Decisions are indexed by $\ell \in [0, \bar{\ell}]$, which measures the loss in surplus incurred if the decision is delegated rather than made by the owner. The loss ℓ is distributed across decisions according to a continuous distribution F with density $f > 0$ on $[0, \bar{\ell}]$. The loss captures, in reduced form, any combination of agent misalignment, inferior information, or inferior judgment relative to the owner. It is not necessary to take a position on its source.

Each decision can be handled in one of two ways.

- **Retention.** The owner makes the decision personally, consuming one unit of attention and incurring no loss.
- **Delegation.** The owner assigns the decision to a subordinate, consuming $s \in (0, 1)$ units of attention for selection, oversight, and review, and incurring the loss ℓ.

The parameter s is the oversight cost of delegation. It summarizes the quality of the owner's monitoring technology, reporting systems, and staff, and, in Section 6, the owner's accumulated delegation capability. The assumption $s < 1$ means that delegation saves attention; the assumption $s > 0$ means that delegation is never free.

A *delegation policy* is a measurable set $D \subseteq [0, \bar{\ell}]$ of decision types that are delegated in every unit. The owner chooses N and D to maximize total surplus

$$\Pi = N \left(V - \int_D \ell \, dF(\ell) \right) \tag{1}$$

subject to the attention constraint

$$N \left(1 - (1 - s) \int_D dF(\ell) \right) \leq T, \tag{2}$$

and the market constraint $N \leq M$. I treat N as continuous throughout.

3.2. The threshold structure

Lemma 1 (Threshold policies). *Any optimal delegation policy is, up to a set of measure zero, an interval $D = [0, \lambda]$ for some $\lambda \in [0, \bar{\ell}]$.*

Lemma 1 is a standard exchange argument: every delegated decision saves the same amount of attention, $1 - s$, so for any given delegation rate the owner should delegate the decisions with the smallest losses. The policy is therefore summarized by a single threshold λ. Define the *delegation rate* $\delta(\lambda) \equiv F(\lambda)$, the *attention requirement per unit*

$$a(\lambda) \equiv 1 - (1 - s)F(\lambda), \tag{3}$$

and the *expected delegation loss per unit*

$$L(\lambda) \equiv \int_0^{\lambda} \ell \, dF(\ell). \tag{4}$$

Note that a is strictly decreasing and L strictly increasing on $[0, \bar{\ell}]$, with $L'(\lambda) = \lambda f(\lambda)$.

3.3. Optimal delegation

When the attention constraint binds, $N = T/a(\lambda)$ and total surplus becomes

$$\Pi(\lambda) = T \cdot \frac{V - L(\lambda)}{a(\lambda)}. \tag{5}$$

The owner's problem is thus to maximize surplus per unit of attention, $\pi(\lambda) \equiv (V - L(\lambda))/a(\lambda)$. Differentiating gives

$$\pi'(\lambda) = \frac{f(\lambda)}{a(\lambda)^2} \, g(\lambda), \qquad g(\lambda) \equiv (1 - s)\big(V - L(\lambda)\big) - \lambda \, a(\lambda). \tag{6}$$

A key property is that $g'(\lambda) = -a(\lambda) < 0$, so g crosses zero at most once and π is strictly quasi-concave. Let λ^u denote the maximizer of π on $[0, \bar{\ell}]$, the *unconstrained-market threshold*. Since $g(0) = (1 - s)V > 0$, we always have $\lambda^u > 0$, and $\lambda^u < \bar{\ell}$ if and only if

$$(1 - s)\big(V - L(\bar{\ell})\big) < s \, \bar{\ell}. \tag{7}$$

I assume (7) throughout, so that it is never optimal to delegate everything.

Proposition 1 (Optimal delegation). *Let $N^u \equiv T/a(\lambda^u)$.*

1. (*Small scale.*) *If $M \leq T$, the unique optimal policy is full control, $\lambda^* = 0$, with $N^* = M$.*
2. (*Attention-constrained scale.*) *If $M \geq N^u$, the optimal threshold is $\lambda^* = \lambda^u$, the unique solution of $g(\lambda) = 0$, and $N^* = N^u$.*
3. (*Intermediate scale.*) *If $T < M < N^u$, the optimal threshold is the unique $\lambda^M \in (0, \lambda^u)$ satisfying $T/a(\lambda^M) = M$, and $N^* = M$.*

In the attention-constrained regime, the optimal threshold satisfies

$$\lambda^u = (1 - s) \cdot \frac{\Pi^u}{T}, \tag{8}$$

where $\Pi^u = \Pi(\lambda^u)$.

Proposition 1 formalizes the regime change that motivates the paper. When the market is small relative to the owner's attention, every decision can be made personally, and since delegation entails a loss with no offsetting gain, full control is optimal. Delegation becomes valuable only when attention binds. A single café can be run by its owner alone; a hundred cannot.

Equation (8) gives the threshold a transparent interpretation. Delegating a marginal decision frees $1 - s$ units of attention, and in the attention-constrained regime each unit of attention is worth Π^u/T, the *average* product of attention, because freed attention is redeployed to operating additional units at the optimal policy. The owner delegates every decision whose loss is smaller than the value of the attention it frees. The threshold is therefore self-referential: owners whose attention is more valuable delegate more, and delegating more is what makes their attention more valuable.

3.4. Comparative statics and the scale ceiling

Proposition 2 (Comparative statics). *In the attention-constrained regime:*

1. λ^u *is strictly decreasing in the oversight cost* s.
2. λ^u *is strictly increasing in unit surplus* V.
3. λ^u *is independent of* T*; scale* N^u *and surplus* Π^u *are proportional to* T.

Lower oversight costs raise delegation through two channels: each delegated decision frees more attention, and the value of each unit of freed attention rises. Higher unit surplus raises delegation because it raises the value of the attention released. Independence from T reflects the constant-returns structure of the model: an owner with more attention operates proportionally more units at the same delegation rate.

Corollary 1 (Scale ceiling). *For any policy,* $N \leq T/a(\lambda) \leq T/s$*. Under full control,* $N \leq T$.

An owner who delegates nothing can operate at most as many units as can be run with full personal attention. An owner who delegates everything is still bounded by the oversight requirement. Holding the oversight technology fixed, the ceiling T/s can only be relaxed by changing the structure of the organization, which is the subject of Section 7.

4. Why Owners Under-Delegate

The model of Section 3 describes an owner who maximizes surplus. I now introduce two departures from this benchmark, motivated by the experimental and psychological evidence reviewed above, and a third, the role of small-scale history, which is developed more fully in Section 6.

4.1. Authority rents and salience

Suppose the owner maximizes the subjective objective

$$W(\lambda) = N\Big[V - (1+\phi)L(\lambda) + b\big(1 - F(\lambda)\big)\Big], \tag{9}$$

subject to the same constraints, where $b \geq 0$ and $\phi \geq 0$.

The parameter b is an *authority rent*: a utility flow the owner derives from each decision made personally, independent of its consequences. It captures the intrinsic value of decision rights (Bartling, Fehr, and Herz, 2014) and, more loosely, the satisfaction of being indispensable. The parameter ϕ is a *salience distortion*: the owner overweights the losses from delegated decisions, which are visible and attributable, relative to the opportunity cost of the attention those decisions would consume, which takes the form of units never operated and is therefore invisible. This is one way of modeling an illusion of control (Langer, 1975): the owner who is involved in every decision prevents small, observable mistakes and does not register the larger losses incurred through forgone scale.

I focus on the attention-constrained regime. Let λ^β denote the threshold chosen by the behavioral owner.

Proposition 3 (Behavioral under-delegation).

1. (*Equivalence.*) *λ^β coincides with the threshold that a surplus-maximizing owner would choose if unit surplus were*
$$\hat{V} \equiv \frac{1}{1+\phi}\left(V - \frac{b\,s}{1-s}\right). \tag{10}$$
2. (*Under-delegation.*) *If $\phi > 0$ or $b > 0$, then $\lambda^\beta < \lambda^u$, and hence the behavioral owner delegates less, operates fewer units, and earns lower surplus than the surplus-maximizing owner.*
3. (*Full control.*) *If $b\,s \geq (1-s)V$, the behavioral owner retains every decision, $\lambda^\beta = 0$, regardless of the market size.*
4. (*Neutrality of the authority rent under costless oversight.*) *As $s \to 0$, the effect of b on λ^β vanishes.*

The equivalence result says that both behavioral distortions act like a tax on the perceived value of the business. The salience distortion scales perceived value down proportionally. The authority rent enters as a subtraction whose size is proportional to $s/(1-s)$, the ratio of the attention a delegated decision still requires to the attention it saves.

Part 4 deserves comment because it is not obvious. The total number of decisions the owner makes personally is

$$R(\lambda) = N\big(1 - F(\lambda)\big) = T \cdot \frac{1 - F(\lambda)}{1 - (1-s)F(\lambda)}. \tag{11}$$

When $s = 0$, $R = T$ for every λ: the attention constraint fixes the number of personally made decisions at the owner's attention endowment, and delegation only determines how many units those decisions are spread over. The owner cannot consume more authority by

delegating less; the only effect of retaining more decisions per unit is to operate fewer units. When $s > 0$, oversight of delegated decisions absorbs attention that would otherwise be spent making decisions, so delegation does reduce the owner's consumption of authority, and the authority rent bites. The implication is that improvements in monitoring and reporting technology attenuate behavioral under-delegation as well as its instrumental cost, which strengthens the case for investment in such systems in organizations whose owners are reluctant to let go.

Two remarks on interpretation are in order. First, if the authority rent is a genuine component of the owner's welfare, under-delegation is privately optimal: the owner consumes part of the return to the business in the form of authority. Proposition 3 then describes a distortion of firm value, not necessarily of owner welfare. The distinction matters for minority shareholders, employees, and creditors, and for the owner's heirs, whose preferences need not coincide with the founder's. Second, the salience distortion is a genuine error. An owner with $\phi > 0$ would, upon reflection, prefer a different policy.

4.2. First-order losses in scale, second-order losses in profit

Proposition 4 (Scale-profit asymmetry). *Suppose f is continuous at λ^u, and let $\lambda = \lambda^u - \Delta$ for small $\Delta > 0$. Then*

$$\ln N(\lambda^u) - \ln N(\lambda) = \frac{(1-s)f(\lambda^u)}{a(\lambda^u)}\,\Delta + o(\Delta), \tag{12}$$

$$\Pi(\lambda^u) - \Pi(\lambda) = \frac{Tf(\lambda^u)}{2\,a(\lambda^u)}\,\Delta^2 + o(\Delta^2). \tag{13}$$

The asymmetry follows from the envelope theorem. At the optimum, the marginal decision's loss exactly offsets the value of the attention it frees, so a small deviation has no first-order effect on surplus. Scale, however, is strictly increasing in the delegation threshold, so the same deviation reduces the number of units operated at a first-order rate.

This result bears on the persistence of under-delegation. An owner who delegates too little operates a smaller organization that is almost as profitable per unit of attention as the optimal one. If owners learn about the quality of their managerial practices from realized profits, the feedback against under-delegation is weak: the profit signal of a moderate error is of the order of its square. In contrast, the owner perceives the losses from each delegated decision directly. The asymmetry between a weak, aggregated signal of the cost of control and a strong, disaggregated signal of the cost of delegation is a natural source of the salience distortion modeled above.

4.3. A numerical illustration

Table 1 illustrates the magnitudes. Losses are uniformly distributed on $[0, 2]$, and $V = T = 1$, so that an owner exercising full control operates one unit and earns surplus of one. Surplus is evaluated at true losses, so the entries measure firm value rather than the owner's subjective utility.

Three features stand out. First, the gain from optimal delegation is substantial: at $s = 0.2$, the surplus-maximizing owner operates two-thirds more units than an owner exercising

Table 1. Optimal and behavioral delegation

s	ϕ	b	Threshold λ	Delegation rate δ	Units N	Surplus Π
0.2	0	0	1.000	0.500	1.667	1.250
0.2	0.5	0	0.607	0.304	1.321	1.199
0.2	0	0.2	0.935	0.467	1.597	1.248
0.2	0.5	0.2	0.572	0.286	1.297	1.191
0.4	0	0	0.667	0.333	1.250	1.111
0.4	0.5	0	0.427	0.214	1.147	1.095
0.4	0	0.2	0.569	0.284	1.206	1.108
0.4	0.5	0.2	0.367	0.183	1.124	1.086

Notes: $\ell \sim U[0,2]$, $V = T = 1$. Thresholds solve the first-order condition in Appendix A; values were confirmed by direct numerical maximization of the subjective objective.

full control and earns 25 percent more. Second, the scale-profit asymmetry of Proposition 4 is visible even for large distortions. With $s = 0.2$ and both distortions active, the owner operates 22 percent fewer units than the optimum but earns less than 5 percent less surplus. Third, the authority rent has a larger effect on the delegation threshold when oversight is costlier: the same $b = 0.2$ lowers the threshold by about 7 percent at $s = 0.2$ and by about 15 percent at $s = 0.4$, consistent with Proposition 3.

4.4. The legacy of small-scale success

Proposition 1 implies that the founder of a business that starts small optimally exercises full control during the early phase. If managerial practices are sticky, whether because of habit, because of the learning frictions implied by Proposition 4, or because delegation is itself a skill, an owner whose business outgrows the small-scale regime enters the attention-constrained regime with practices calibrated to conditions that no longer apply. The practice that delivered early success then becomes the binding constraint on later growth. Section 6 formalizes the case in which the friction is an accumulable capability.

5. Delegating Objectives versus Prescribing Solutions

The baseline model treats delegation as a binary choice. In practice, an owner can assign a decision to a subordinate while retaining control over its content, by specifying the action the subordinate should take. This section shows that such *prescriptive* delegation is, in the relevant cases, dominated, and that an owner who habitually prescribes operates as if oversight were more costly than it is.

Suppose each decision can be handled in one of three modes. Retention and delegation are as before. Under *prescription*, the owner formulates the solution and assigns only its execution to the subordinate. Formulating the solution requires $c \in (0, 1-s)$ units of attention in addition to the oversight cost s, and the decision incurs a loss $\iota \geq 0$ that reflects the subordinate's specific knowledge left unused, in the sense of Jensen and Meckling (1992): the owner's solution is based on the owner's information, not the subordinate's. Delegation of the objective, by contrast, incurs the loss ℓ, which now reflects only misalignment and errors of judgment.

Let μ denote the shadow value of the owner's attention, which by Proposition 1 equals Π^u/T in the attention-constrained regime. The total cost of each mode is μ for retention, $\mu s + \ell$ for delegation, and $\mu(s+c) + \iota$ for prescription.

Proposition 5 (Prescription).

1. *Prescription is strictly preferred to delegation only if $\ell > \iota + \mu c$.*
2. *For decisions in which the subordinate has expertise, in the sense that the information loss from prescription is at least as large as the loss from delegation ($\iota \geq \ell$), prescription is strictly dominated by delegation.*
3. *An owner who prescribes a fraction $p \in (0,1]$ of the decisions that would optimally be delegated faces an effective oversight cost $s + p\,c$. By Proposition 2, such an owner chooses a lower delegation threshold and operates fewer units.*
4. *As $c \to 1 - s$, the attention saved by prescription vanishes and prescription becomes equivalent to retention.*

The proposition makes precise a familiar managerial observation. Stating the need and leaving the solution to the expert transfers the cognitive burden of the decision. Dictating the solution does not. The owner who dictates continues to bear the cost of working out what should be done, and the organization loses the subordinate's local knowledge. In the limiting case in which formulating a solution is as costly as making the decision, prescriptive delegation is delegation in name only: the owner remains the decision maker, and the subordinate supplies execution.

The static model understates the cost of prescription. In the framework of Aghion and Tirole (1997), an agent whose recommendations are routinely overridden has little incentive to acquire the information on which good recommendations depend. Habitual prescription therefore erodes the subordinate's specific knowledge and raises the losses from future delegation, which shifts the distribution F to the right over time. Prescription is self-reinforcing in the same way as the dynamics analyzed next.

6. Delegation as Capital

The oversight cost s has so far been treated as a fixed parameter. It is more natural to regard it as the product of experience. Selecting the right people, specifying objectives, designing reporting, and calibrating trust are skills, and like other skills they improve with practice and decay with disuse. This section formalizes that idea and shows that it generates path dependence.

6.1. Dynamics

Time is discrete. The owner's *delegation capital* $K_t \geq 0$ determines the oversight cost $s_t = \sigma(K_t)$, where σ is continuous and strictly decreasing, with $\sigma(0) = \bar{s} < 1$ and $\lim_{K\to\infty} \sigma(K) = \underline{s} > 0$. Capital accumulates through use and depreciates at rate $\rho \in (0,1)$:

$$K_{t+1} = (1-\rho)K_t + \delta_t, \tag{14}$$

where δ_t is the delegation rate in period t. This is a learning-by-doing specification (Arrow, 1962) applied to the managerial function itself.

I first consider a *myopic* owner who, in every period, chooses the static optimum given the current oversight cost: $\delta_t = \delta^*(\sigma(K_t))$, where $\delta^*(s) = F(\lambda^u(s))$ in the attention-constrained regime. Define the map $\Phi(K) \equiv \delta^*(\sigma(K))$. By Proposition 2 and the monotonicity of σ, Φ is continuous and increasing. Steady states solve

$$\rho K = \Phi(K). \tag{15}$$

Proposition 6 (Path dependence).

1. *At least one steady state exists, and every steady state lies in* $(0, 1/\rho]$.
2. *If* Φ *is sufficiently convex over some range, there may be multiple steady states. Generically, steady states alternate between locally stable* (*at which* Φ *crosses* ρK *from above*) *and unstable* (*at which it crosses from below*).
3. *If there are several steady states, let* $K^\dagger$ *be the smallest unstable one and* K_L *the smallest steady state. A myopic owner with* $K_0 < K^\dagger$ *converges to* K_L.
4. *An owner whose initial phase was spent in the small-scale regime of Proposition 1 chooses* $\delta_t = 0$ *during that phase and therefore enters the attention-constrained regime with depreciated capital. If that phase was long enough, the owner converges to* K_L.
5. *A temporary shock that raises* K *above* $K^\dagger$, *such as a period during which the owner is unable to make decisions personally, moves the owner permanently to a higher steady state.*

The convexity required in part 2 arises naturally when the returns to delegation experience are initially low and then accelerate, for example because the first delegated decisions are the hardest to monitor and the infrastructure of delegation (reporting systems, a trusted management team, routines for escalation) has fixed costs that are only recovered at scale.

6.2. An example

Let losses again be uniform on $[0, 2]$, with $V = T = 1$, and let $\sigma(K) = 0.15 + 0.75/(1 + K^4)$ and $\rho = 0.25$. The system has three steady states, at $K \approx 0.20$, 0.99, and 2.04. The outer two are stable; the middle one separates their basins of attraction.

At the low steady state, the oversight cost is approximately 0.90, the owner delegates about 5 percent of decisions, operates about 1.01 units, and earns surplus of about 1.00, which is virtually indistinguishable from full control. At the high steady state, the oversight cost is approximately 0.19, the owner delegates about 51 percent of decisions, operates 1.70 units, and earns surplus of about 1.26. The two owners are identical in their preferences, their attention, their markets, and the technology available to them. They differ only in their histories.

An owner starting with $K_0 = 0$, the natural initial condition for a founder who has spent the formative years of the business in the small-scale regime, converges to the low steady state. A one-time increase in delegation sufficient to raise K above 0.99 moves the same owner permanently to the high steady state.

6.3. Forward-looking owners and awareness

A forward-looking owner recognizes that delegation today lowers oversight costs tomorrow. Because surplus per unit of attention is decreasing in s (by the envelope theorem, $d\pi^*/ds = -(V - L(\lambda^u))F(\lambda^u)/a(\lambda^u)^2 < 0$), the continuation value is increasing in K, and a forward-looking owner has an additional motive to delegate beyond the static optimum. When the owner is sufficiently patient, this investment motive can eliminate the low steady state.

The relevant distinction is therefore not between owners who are temperamentally inclined to delegate and owners who are not, but between owners who perceive delegation capability as an asset to be built and owners who do not. Delegation capability behaves like a muscle: it develops with use and atrophies without it. The binding condition for escaping the trap is awareness of the owner's own capacity constraint, which is precisely what the salience distortion of Section 4 and the weak profit feedback of Proposition 4 suppress.

7. Holdings as Recursive Delegation

The ceiling $N \leq T/s$ of Corollary 1 binds any owner who oversees operating units directly. Business groups relax it by interposing layers of managers, each of whom faces a delegation problem of the same form. This section shows that the returns to delegation then compound.

7.1. A recursive hierarchy

Let tier-0 nodes be operating units with value $v_0 = V$. A tier-k node, for $k = 1, \ldots, K$, is a manager endowed with attention T who oversees n_k tier-$(k-1)$ nodes. Each subordinate node generates a unit mass of decisions requiring the tier-k manager's involvement: appointments, budgets, capital allocation, and the resolution of exceptions. If a decision of type ℓ is delegated to the subordinate, the loss is $\ell\, v_{k-1}$; that is, the stakes of decisions scale with the value of the node they concern. Losses are distributed according to F as before. The owner sits at tier K. Given threshold λ_k at tier k, the attention constraint binds with $n_k = T/a(\lambda_k)$, and

$$v_k = n_k\, v_{k-1}\big(1 - L(\lambda_k)\big) = v_{k-1}\,\gamma(\lambda_k), \qquad \gamma(\lambda) \equiv \frac{T\big(1 - L(\lambda)\big)}{a(\lambda)}. \tag{16}$$

Proposition 7 (Compounding).

1. *The value of the group is $v_K = V \prod_{k=1}^{K} \gamma(\lambda_k)$, and the optimal threshold is the same at every tier, $\lambda_k = \lambda^\circ$, where λ° is the solution of the baseline problem with $V = 1$.*

2. *Relative to a group in which every tier exercises full control, the gain from optimal delegation is $\big[(1 - L(\lambda^\circ))/a(\lambda^\circ)\big]^K$, which grows geometrically in the number of tiers.*

3. *If every tier is subject to the same behavioral distortion and chooses $\lambda^\beta < \lambda^\circ$, the group's value relative to the optimum is $\big[\gamma(\lambda^\beta)/\gamma(\lambda^\circ)\big]^K$, which declines geometrically in the number of tiers.*

The multiplicative structure follows from the scale invariance of decision stakes: each tier solves the same normalized problem, and its delegation policy multiplies the value of everything below it. Two implications deserve emphasis.

First, the work of the owner at the apex of a group is almost entirely concerned with people. In the recursive structure, the decisions that reach the owner are decisions about

tier-$(K-1)$ managers, not about operations. The owner's role resembles that of a conductor who plays no instrument: the music is produced entirely by others, and the conductor's contribution lies in selection, coordination, and the decision of when to intervene.

Second, a culture of under-delegation is far more costly in a large group than in a single firm. Using the parameters of Table 1 with $s = 0.2$, $\phi = 0.5$, and $b = 0.2$ applied at every tier (with b measured in units of subordinate value), the loss relative to the optimum is 4.7 percent with one tier, 9.3 percent with two, 13.6 percent with three, and 17.7 percent with four. A distortion that is a minor inefficiency in a single firm becomes a first-order determinant of value in a deep group. The owner's own delegation behavior matters doubly here, because a non-delegating owner is unlikely to select, reward, or retain subordinates who delegate.

7.2. Remarks

The frictionless version above abstracts from managerial compensation. If each intermediate manager receives a wage w, adding a tier to a group of value v is worthwhile only if $(\gamma(\lambda^\circ) - 1)\, v > w \cdot (\text{number of new managers})$. Since γ is increasing in the delegation threshold, under-delegating groups are not only less valuable at a given depth but are also optimally shallower. This provides one interpretation of the observation that groups with comparable assets differ greatly in organizational depth.

The scale invariance of decision stakes is the assumption that delivers exact compounding. If stakes grow less than proportionally with node value, for example because higher tiers make more standardized decisions, the effect of delegation at higher tiers is attenuated, but the qualitative conclusion that delegation at each tier multiplies the value of the structure below it is preserved.

8. A Diagnostic: Decisions No Longer Made

The model suggests that the number of units an owner controls is an incomplete and potentially misleading indicator of organizational capacity. This section derives a simple accounting identity that makes the point precise and suggests an observable alternative.

Allow the owner to operate more units than the attention constraint permits, $N > T/a(\lambda)$. Some decisions are then neither made by the owner nor delegated: they are *neglected*, resolved by default, by inertia, or by whoever happens to be present, without either the owner's judgment or an explicit assignment of authority. For each unit, let δ be the fraction of decisions delegated, m the fraction made personally by the owner, and ν the fraction neglected, so that $\delta + m + \nu = 1$. Attention requires $N(m + s\delta) \leq T$.

Proposition 8 (Accounting). *For any $N > 0$,*

$$\nu \;\geq\; 1 - (1 - s)\,\delta - \frac{T}{N}. \tag{17}$$

Consequently:

1. *Zero neglect requires a delegation rate of at least*

$$\delta_{\min}(N) = \frac{1 - T/N}{1 - s}. \tag{18}$$

2. *The number of decisions the owner no longer makes, $N\delta$, must satisfy*

$$N\delta \;\geq\; \frac{N - T}{1 - s}, \tag{19}$$

   which increases in N with slope $1/(1-s) > 1$.

3. *If N rises while δ is held fixed, the lower bound on the neglect rate ν rises.*

Part 2 is the central diagnostic. Each additional unit in the portfolio must be accompanied by more than one additional unit of decisions that the owner has stopped making, because delegated decisions still absorb oversight. With $s = 0.2$ and $T = 1$, an owner operating two units must delegate at least 62.5 percent of decisions to avoid neglect, an owner operating three units at least 83 percent, and no owner can operate more than five.

Part 3 describes the pathology of the owner who acquires companies without changing the way they are run. Such an owner does not operate a holding in any meaningful organizational sense; the owner operates a large single-person organization, in which a rising share of decisions is made by no one. The natural diagnostic question is therefore not how many companies a person controls, but how many decisions that person no longer makes. When the first number rises and the second does not, the identity (17) implies that the difference is absorbed by neglect.

9. Empirical Implications and Measurement

The model generates several predictions that are, in principle, testable with existing types of data.

P1 (Monitoring technology). Holding firm size and sector fixed, delegation rates should be higher where the oversight cost of delegation is lower, for instance after the adoption of reporting, workflow, or enterprise software that makes delegated decisions observable at low cost (Proposition 2).

P2 (Interaction of authority rents and oversight costs). Proxies for owners' intrinsic preference for control should predict under-delegation more strongly where oversight is costly, and the predictive power should decline after improvements in monitoring technology (Proposition 3, part 4). This interaction distinguishes the authority-rent channel from purely instrumental explanations of under-delegation, which predict no such pattern.

P3 (Hysteresis). Conditional on current size and fundamentals, owners who spent longer in the small-scale regime should delegate less. Exogenous temporary absences of the owner, for example due to illness or extended travel, should be followed by persistent increases in delegation and in scale (Proposition 6).

P4 (Scale-profit asymmetry). Cross-sectional dispersion in delegation should be associated with substantially larger dispersion in scale than in surplus per unit of owner attention (Proposition 4).

P5 (Depth of business groups). Within business groups, measures of delegation at intermediate tiers should be positively associated with organizational depth, and the value consequences of delegation should be larger in deeper groups (Proposition 7).

Measurement. The key primitives of the model can be measured more directly than is often assumed. Time-use surveys and diaries of chief executives (Bandiera et al., 2018, 2020) measure the allocation of owner attention. Survey instruments on the decentralization of decision rights (Bloom, Sadun, and Van Reenen, 2012) measure the delegation rate δ for specific classes of decisions. Increasingly, the decision flow itself is recorded: the task-tracking, request, and approval systems used by multi-establishment firms log who initiated each request, who resolved it, and how often it was escalated. Such records permit direct measurement of the three quantities in identity (17): decisions retained by the owner, decisions delegated, and decisions left unresolved. They also permit estimation of the oversight cost s, as the owner's time spent reviewing delegated decisions relative to the time required to make them.

Identification. The principal empirical challenge is that delegation and scale are jointly determined. The model suggests sources of variation that shift the oversight cost without directly affecting demand, such as the staggered adoption of monitoring technology, and sources that shift delegation capital without directly affecting current fundamentals, such as exogenous owner absences.

10. Conclusion

This paper has studied delegation as the allocation of a fixed stock of attention across a continuum of decisions. The central object is not the number of decisions an owner makes but the number the owner no longer makes. The model formalizes several observations that are familiar from practice but have not been brought together in a single framework: that full control is optimal at small scale and becomes the binding constraint at large scale; that a preference for authority and an overweighting of visible delegation errors act as a tax on the perceived value of the business; that under-delegation is costly for scale but nearly costless for profit, and is therefore weakly disciplined by the owner's own experience; that delegation capability is accumulated through use, so that small-scale histories can produce persistent traps; and that in multi-tier organizations the returns to delegation compound, so that cultures of control are most costly where organizations are deepest.

Several extensions are natural. The units in the model are homogeneous and the distribution of delegation losses is exogenous. Allowing the owner to invest in the quality of subordinates, and thereby to shift F, would connect the model to the literature on managerial talent and to the endogenous formation of management teams. The attention endowment is fixed; in practice, owners can expand it through staff, deputies, or co-owners, each of whom raises the issues analyzed here at one remove. The dynamic analysis considers a myopic owner and characterizes the forward-looking case only qualitatively. A full dynamic

treatment, together with heterogeneity in owner preferences, would allow the model to be taken to firm-level data on growth and organizational structure.

The broader implication is that the scalability of an organization is, to a significant extent, a property of its owner's decision rules rather than of its assets. An owner who controls many companies but makes the same decisions in each of them has extended the reach of a single person; an owner who has stopped making most of those decisions has built an organization.

Appendix A. Proofs

Proof of Lemma 1. Fix any measurable delegation set D with F-measure $\delta = \int_D dF$. The attention requirement per unit, $1-(1-s)\delta$, depends on D only through δ. Let λ satisfy $F(\lambda) = \delta$ and set $D' = [0, \lambda]$. For any D with the same measure, $\int_{D\setminus D'} dF = \int_{D'\setminus D} dF$, and every $\ell \in D \setminus D'$ satisfies $\ell \geq \lambda$ while every $\ell \in D' \setminus D$ satisfies $\ell \leq \lambda$. Hence $\int_D \ell\, dF \geq \int_{D'} \ell\, dF$, with strict inequality whenever D and D' differ on a set of positive measure, since $f > 0$. Replacing D with D' therefore weakly raises surplus for given N and leaves the attention constraint unchanged. □

Proof of Proposition 1. By Lemma 1, the owner chooses (N, λ) to maximize $N(V - L(\lambda))$ subject to $Na(\lambda) \leq T$ and $N \leq M$.

Part 1. Since $L \geq 0$ and $N \leq M$, surplus is at most MV. If $M \leq T$, the pair $(N, \lambda) = (M, 0)$ is feasible because $a(0) = 1$, and attains MV. Any $\lambda > 0$ yields $L(\lambda) > 0$ and therefore surplus strictly below MV for every feasible N.

Parts 2 and 3. Let $M > T$. At an optimum, at least one constraint binds; otherwise N could be increased. If $N < M$, the attention constraint binds and $N = T/a(\lambda)$, so surplus is $\Pi(\lambda) = T\pi(\lambda)$. Differentiating $\pi(\lambda) = (V - L(\lambda))/a(\lambda)$, using $L' = \lambda f$ and $a' = -(1-s)f$, gives

$$\pi'(\lambda) = \frac{-\lambda f(\lambda)\, a(\lambda) + (1-s) f(\lambda)\big(V - L(\lambda)\big)}{a(\lambda)^2} = \frac{f(\lambda)}{a(\lambda)^2}\, g(\lambda).$$

Moreover, $g'(\lambda) = -(1-s)\lambda f(\lambda) - a(\lambda) + (1-s)\lambda f(\lambda) = -a(\lambda) < 0$. Since $g(0) = (1-s)V > 0$ and, under (7), $g(\bar{\ell}) = (1-s)(V - L(\bar{\ell})) - s\bar{\ell} < 0$, g has a unique root $\lambda^u \in (0, \bar{\ell})$, at which π attains its unique maximum; π is strictly increasing on $[0, \lambda^u]$ and strictly decreasing on $[\lambda^u, \bar{\ell}]$.

The scale $T/a(\lambda)$ is strictly increasing in λ. If $T/a(\lambda^u) \leq M$, the policy (N^u, λ^u) is feasible and attains the maximum of $T\pi(\lambda)$, which is an upper bound on surplus whenever the attention constraint binds; any policy at which only the market constraint binds is dominated, because with slack attention at $N = M$ the owner's surplus $M(V - L(\lambda))$ is at most $T\pi(\lambda) \cdot Ma(\lambda)/T < T\pi(\lambda) \leq T\pi(\lambda^u)$. This establishes Part 2.

If $T < M < T/a(\lambda^u)$, there is a unique $\lambda^M \in (0, \lambda^u)$ with $T/a(\lambda^M) = M$. For $\lambda \leq \lambda^M$, the attention constraint binds and surplus $T\pi(\lambda)$ is increasing in λ. For $\lambda > \lambda^M$, the market constraint binds, $N = M$, and surplus $M(V - L(\lambda))$ is strictly decreasing in λ. Hence λ^M is optimal. This establishes Part 3.

Equation (8). At λ^u, $g(\lambda^u) = 0$ implies $\lambda^u = (1-s)(V - L(\lambda^u))/a(\lambda^u) = (1-s)\pi(\lambda^u) = (1-s)\Pi^u/T$. □

Proof of Proposition 2. The threshold λ^u is defined by $g(\lambda; s, V) = 0$ with $g_\lambda = -a < 0$. Using $\partial a/\partial s = F(\lambda)$,

$$\frac{\partial g}{\partial s} = -\big(V - L(\lambda)\big) - \lambda F(\lambda) < 0, \qquad \frac{\partial g}{\partial V} = 1 - s > 0.$$

By the implicit function theorem, $d\lambda^u/ds = -g_s/g_\lambda < 0$ and $d\lambda^u/dV = -g_V/g_\lambda > 0$. Since T does not enter g, λ^u is independent of T, and $N^u = T/a(\lambda^u)$ and $\Pi^u = T\pi(\lambda^u)$ are proportional to T. □

Proof of Corollary 1. The attention constraint implies $N \le T/a(\lambda)$, and $a(\lambda) \ge a(\bar{\ell}) = s$. Under full control, $a(0) = 1$. □

Proof of Proposition 3. In the attention-constrained regime, the behavioral owner maximizes $T\,U(\lambda)/a(\lambda)$ with $U(\lambda) = V - (1+\phi)L(\lambda) + b(1 - F(\lambda))$. Since $U'(\lambda) = -[(1+\phi)\lambda + b]f(\lambda)$, the derivative of U/a has the sign of

$$-\big[(1+\phi)\lambda + b\big]a(\lambda) + (1-s)U(\lambda).$$

Expanding and using $(1-s)(1-F) - a = -s$,

$$(1-s)U(\lambda) - b\,a(\lambda) - (1+\phi)\lambda a(\lambda) = (1-s)\big[V - (1+\phi)L(\lambda)\big] - bs - (1+\phi)\lambda a(\lambda) \equiv h(\lambda).$$

Dividing by $1+\phi$,

$$\frac{h(\lambda)}{1+\phi} = (1-s)\big[\hat{V} - L(\lambda)\big] - \lambda a(\lambda),$$

with $\hat{V}$ as in (10). This is the function g with V replaced by $\hat{V}$, which proves Part 1. As in the proof of Proposition 1, $h' = -(1+\phi)a < 0$.

For Part 2, evaluate h at λ^u. Since $g(\lambda^u) = 0$, $(1-s)L(\lambda^u) + \lambda^u a(\lambda^u) = (1-s)V$, hence

$$h(\lambda^u) = (1-s)V - (1+\phi)(1-s)V - bs = -\phi(1-s)V - bs,$$

which is strictly negative if $\phi > 0$ or $b > 0$ (recall $s > 0$). Since h is strictly decreasing, $\lambda^\beta < \lambda^u$. Because π is strictly increasing on $[0, \lambda^u]$ and $N = T/a(\lambda)$ is strictly increasing, both true surplus and scale are strictly lower at λ^β.

For Part 3, $h(0) = (1+\phi)(1-s)\hat{V}$, which is non-positive if and only if $bs \ge (1-s)V$. In that case $h < 0$ on $(0, \bar{\ell}]$, so the subjective objective is strictly decreasing and $\lambda^\beta = 0$. Since the subjective objective at $\lambda = 0$ is independent of M whenever $M \ge T$, and full control is optimal for $M \le T$ by Proposition 1, the conclusion holds for every market size.

For Part 4, $\hat{V} \to V/(1+\phi)$ as $s \to 0$, which does not depend on b. By continuity of the root of h in $(s, \hat{V})$, the effect of b on λ^β vanishes. Equation (11) follows from $N = T/a(\lambda)$. □

Proof of Proposition 4. For (12), $\ln N(\lambda) = \ln T - \ln a(\lambda)$, so $d\ln N/d\lambda = (1-s)f(\lambda)/a(\lambda)$, which is continuous at λ^u; a first-order expansion gives (12). For (13), write

$$\Pi(\lambda^u) - \Pi(\lambda^u - \Delta) = T\int_{\lambda^u - \Delta}^{\lambda^u} \frac{f(x)}{a(x)^2}\, g(x)\, dx.$$

Since $g(\lambda^u) = 0$ and g is continuously differentiable with $g' = -a$, $g(x) = a(\lambda^u)(\lambda^u - x) + o(\lambda^u - x)$. By continuity of f and a at λ^u, the integrand equals $\frac{f(\lambda^u)}{a(\lambda^u)}(\lambda^u - x) + o(\lambda^u - x)$. Integrating gives $\frac{Tf(\lambda^u)}{2a(\lambda^u)}\Delta^2 + o(\Delta^2)$. □

Proof of Proposition 5. Parts 1 and 2 follow from comparing $\mu(s+c) + \iota$ with $\mu s + \ell$: prescription is strictly cheaper if and only if $\ell > \iota + \mu c$, which is impossible when $\iota \geq \ell$ because $\mu c > 0$. For Part 3, if a fraction p of delegated decisions are prescribed, each delegated decision consumes on average $s + pc$ units of attention, so the attention requirement per unit is $1 - (1 - s - pc)F(\lambda)$. This is the baseline problem with oversight cost $s + pc$ and losses weakly increased (if $\iota \geq \ell$ on the prescribed set), and Proposition 2 applies. Part 4 is immediate since prescription saves $1 - s - c$ units of attention relative to retention. □

Proof of Proposition 6. Part 1. Φ is continuous on $[0, \infty)$ with values in $[0, 1]$. Define $G(K) = \Phi(K) - \rho K$. Then $G(0) = \Phi(0) = \delta^*(\bar{s}) > 0$, since $\lambda^u > 0$ for any $s < 1$, and $G(1/\rho) = \Phi(1/\rho) - 1 \leq 0$. By the intermediate value theorem, G has a root in $(0, 1/\rho]$. For $K > 1/\rho$, $G(K) < 0$, so no steady state exceeds $1/\rho$.

Part 2. The dynamics are $K_{t+1} = (1-\rho)K_t + \Phi(K_t) \equiv H(K_t)$, with H continuous and strictly increasing, since Φ is increasing. A one-dimensional system with an increasing map converges monotonically from any initial condition to a fixed point. A fixed point at which G changes sign from positive to negative attracts initial conditions in a neighborhood, and one at which G changes sign from negative to positive repels them. Since $G(0) > 0$ and $G(K) < 0$ for $K > 1/\rho$, the number of sign changes is odd, and for generic parameters stable and unstable steady states alternate, beginning and ending with stable ones. Multiplicity requires $\Phi'(K) > \rho$ over some range, that is, sufficient curvature of Φ.

Part 3. On $[0, K_L)$, $G > 0$, so K_t increases monotonically toward K_L. On $(K_L, K^\dagger)$, by the definition of $K^\dagger$ as the smallest unstable steady state and the alternation of steady states, $G < 0$, so K_t decreases monotonically toward K_L. Since H is increasing, K_t remains on the same side of each fixed point, and convergence to K_L follows.

Part 4. In the small-scale regime, $\delta_t = 0$ by Proposition 1, so $K_{t+\tau} = (1-\rho)^\tau K_t \to 0$. For a sufficiently long small-scale phase, the owner enters the attention-constrained regime with $K < K^\dagger$, and Part 3 applies.

Part 5. If a shock raises K above $K^\dagger$ and $K^\dagger$ is unstable, K_t thereafter converges to the next stable steady state above $K^\dagger$. □

Proof of Proposition 7. Equation (16) gives $v_K = V \prod_k \gamma(\lambda_k)$. Since each factor depends only on its own threshold and all factors are positive, the product is maximized by maximizing each factor separately. Each factor equals $T(1 - L(\lambda))/a(\lambda)$, which is T times the baseline objective π with $V = 1$, so the maximizer is λ° at every tier. Under full control, $\gamma(0) = T$, so the ratio of optimal to full-control value is $[\gamma(\lambda^\circ)/T]^K = [(1 - L(\lambda^\circ))/a(\lambda^\circ)]^K$. Part 3 follows directly from the product form. □

Proof of Proposition 8. The attention constraint $N(m + s\delta) \leq T$ implies $m \leq T/N - s\delta$. Substituting $m = 1 - \delta - \nu$ gives $1 - \delta - \nu \leq T/N - s\delta$, which rearranges to (17). Setting $\nu = 0$ gives (18), and multiplying (18) by N gives (19). Part 3 follows because the right-hand side of (17) is strictly increasing in N. □

Appendix B. Numerical Details

All illustrations assume $\ell \sim U[0, \bar{\ell}]$ with $\bar{\ell} = 2$, so that $F(\lambda) = \lambda/2$ and $L(\lambda) = \lambda^2/4$ on $[0, 2]$, with $V = T = 1$. Condition (7) becomes $(1 - s)(1 - 1) < 2s$, which holds for every $s > 0$.

Table 1. For each (s, ϕ, b), the threshold is the unique root of $h(\lambda) = (1 - s)[1 - (1 + \phi)\lambda^2/4] - bs - (1 + \phi)\lambda[1 - (1 - s)\lambda/2]$ on $[0, 2]$, computed by bracketing. Each root was confirmed by direct numerical maximization of the subjective objective $U(\lambda)/a(\lambda)$. Reported surplus is $\Pi = (1 - \lambda^2/4)/a(\lambda)$, evaluated at true losses. For the surplus-maximizing owner, $g(\lambda) = (1 - s) - \lambda + (1 - s)\lambda^2/4$, so the threshold has the closed form $\lambda^u = 2\big[1 - \sqrt{1 - (1 - s)^2}\,\big]/(1 - s)$. This gives $\lambda^u = 1$ at $s = 0.2$ and $\lambda^u = 2/3$ at $s = 0.4$, consistent with equation (8): $0.8 \times 1.25 = 1$ and $0.6 \times 1.111 = 0.667$.

Section 6.2. With $\sigma(K) = 0.15 + 0.75/(1 + K^4)$ and $\rho = 0.25$, steady states are the roots of $\delta^*(\sigma(K)) - 0.25K$, located on a grid of 40,001 points on $[0, 4]$ and refined numerically. Simulating the map $K_{t+1} = 0.75K_t + \delta^*(\sigma(K_t))$ for 200 periods from $K_0 \in \{0, 0.5\}$ converges to $K \approx 0.203$, and from $K_0 \in \{1.5, 3\}$ to $K \approx 2.037$. At the low steady state, $s \approx 0.899$, $\lambda \approx 0.102$, $\delta \approx 0.051$, $N \approx 1.005$, $\Pi \approx 1.003$. At the high steady state, $s \approx 0.191$, $\lambda \approx 1.019$, $\delta \approx 0.509$, $N \approx 1.701$, $\Pi \approx 1.259$.

Section 7.1. For $s = 0.2$, $\gamma(\lambda^\circ) = 1.25$ and, under $\phi = 0.5$ and $b = 0.2$ at every tier, $\gamma(\lambda^\beta) \approx 1.1907$. The ratio is approximately 0.9526, so relative values at depths one through four are approximately 0.953, 0.907, 0.864, and 0.823.

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